

D893

Linear Algebra

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A data analyst at a streaming company stores a 3×2 matrix A of viewer scores and a 2×4 matrix B of genre weights. The company has 12 million subscribers. What is the size of AB ?

- A. 3×4
- B. 4×3
- C. AB is undefined.
- D. 2×2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A 3×3 matrix A used in a computer graphics class has $\det(A) = 5$. The class meets in a lab with dual monitors. What is $\det(2A)$?

- A. 10
- B. 40
- C. 20
- D. $5/2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A square matrix A has $\det(A) = 0$. What must be true?

- A. A is the zero matrix.
- B. $Ax = 0$ has a nonzero solution.
- C. A is invertible.
- D. $Ax = b$ has exactly one solution for every b .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student in an online study group row-reduces an augmented matrix to $\begin{bmatrix} 1 & 2 & 0 & | & 3 \\ 0 & 0 & 1 & | & 4 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$. The study group meets on Sunday evenings. How many solutions does the system have?

- A. No solution, because of the row of zeros.
- B. Exactly one solution
- C. Infinitely many
- D. Exactly three solutions

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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5. [TRANSFER — reversed stem / different variable]

A reduced augmented matrix has the last row $[0 \ 0 \ 0 \ | \ 5]$. How many solutions does the system have?

- A. Infinitely many
- B. None
- C. Exactly one
- D. Exactly five

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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6.

An engineer at a bridge firm has four force vectors in $\mathbb{R}^3$. The firm has won two design awards. None of the vectors is a multiple of another. What must be true?

- A. They are linearly dependent.
- B. They are linearly independent, because none is a multiple of another.
- C. They must be a basis for $\mathbb{R}^3$.
- D. You cannot tell without row reducing.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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7.

A linear transformation T in a robotics course sends e_1 to $(2, 1)$ and e_2 to $(-1, 3)$. The course uses a robot kit with six motors. What is $T(3, 2)$?

- A. $(6, 6)$
- B. $(8, 3)$
- C. $(1, 4)$
- D. $(4, 9)$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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8.

A population model in an ecology study uses $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. The study tracks a bird species on a coastal island. What are the eigenvalues of A ?

- A. 5 and 2
- B. -5 and -2
- C. 7 and 10
- D. 4 and 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A 2×2 matrix has eigenvalues 3 and -2. What is its determinant?

- A. 5
- B. 6
- C. -6
- D. 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher fits a line to 20 data points from a class survey on sleep and grades. The survey was done during finals week. The system $Ax = b$ has no exact solution. Which equation gives the least-squares solution?

- A. $AA^T x = b$
- B. Solve $Ax = b$ exactly by row reduction.
- C. $A^T A x = A^T b$
- D. $x = A^{-1} b$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 5×7 matrix from a supply-chain model has rank 3. The model covers 14 warehouses. What is the dimension of its null space?

- A. 11
- B. 3
- C. 2
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A 6×8 matrix has a null space of dimension 5. What is its rank?

- A.** 5
- B.** 1
- C.** 13
- D.** 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A data analyst at a streaming company stores a 3×2 matrix A of viewer scores and a 2×4 matrix B of genre weights. ~~The company has 12 million subscribers.~~ What is the size of AB?

Target word: size of product **Decides:** Outer dimensions give the result.

A	Correct	$(3 \times 2)(2 \times 4)$ gives 3×4 .
B	Inverted operation	That would be the size of $(AB)^T$.
C	Confusion set	The inner sizes match (2 and 2); it is BA that is undefined.
D	Inverted operation	Uses the inner dimensions.

2. Answer B Concept 2

A 3×3 matrix A used in a computer graphics class has $\det(A) = 5$. ~~The class meets in a lab with dual monitors.~~ What is $\det(2A)$?

Target word: det of a scaled matrix **Decides:** Each of 3 rows is scaled by 2.

A	Dropped element	Forgets the power $n = 3$.
B	Correct	$2^3 \times 5 = 40$.
C	Wrong stem number	Uses 2^2 as if the matrix were 2×2 .
D	Inverted operation	Divides instead of multiplying.

3. Answer B Concept 2 · transfer

A square matrix A has $\det(A) = 0$. What must be true?

Target word: consequence of zero determinant **Decides:** Singular matrix.

A	Absolute language	Many nonzero matrices have determinant 0.
B	Correct	$\det = 0$ means A is not invertible, so its null space is nontrivial.
C	Inverted operation	A zero determinant means not invertible.
D	Confusion set	That holds only when $\det$ is not 0.

4. Answer C Concept 3

A student ~~in an online study group~~ row-reduces an augmented matrix to $[1 \ 2 \ 0 \mid 3; 0 \ 0 \ 1 \mid 4; 0 \ 0 \ 0 \mid 0]$. ~~The study group meets on Sunday evenings.~~ How many solutions does the system have?

Target word: number of solutions **Decides:** Consistent with x_2 free.

A	Confusion set	A zero row with a 0 on the right is $0 = 0$, which is fine.
B	Dropped element	Ignores the free variable x_2 .
C	Correct	x_2 is free; $x_1 = 3 - 2x_2$ and $x_3 = 4$.
D	Wrong stem number	Counts the rows, not the solutions.

5. Answer B Concept 3 · transfer

A reduced augmented matrix has the last row $[0 \ 0 \ 0 \ | \ 5]$. How many solutions does the system have?

Target word: number of solutions **Decides:** $0 = 5$ is impossible.

A	Confusion set	Confuses this with a zero row that is consistent.
B	Correct	The row says $0 = 5$.
C	Adjacent category	Depends on pivots only if the system is consistent.
D	Wrong stem number	Reads the 5 as a count.

6. Answer A Concept 4

An engineer at a bridge firm has four force vectors in $\mathbb{R}^3$. ~~The firm has won two design awards.~~ None of the vectors is a multiple of another. What must be true?

Target word: independence **Decides:** Four vectors in a 3-dimensional space.

A	Correct	More than 3 vectors in $\mathbb{R}^3$ are always dependent.
B	Confusion set	Pairwise non-multiples can still be dependent.
C	Inverted operation	A basis must be independent; four vectors cannot be.
D	Adjacent category	The count alone settles dependence here.

7. Answer D Concept 5

A linear transformation T in a robotics course sends e_1 to $(2, 1)$ and e_2 to $(-1, 3)$. ~~The course uses a robot kit with six motors.~~ What is $T(3, 2)$?

Target word: apply T **Decides:** Columns of the matrix are $T(e_1)$ and $T(e_2)$.

A	Wrong stem number	Multiplies 3 and 2 by the wrong entries.
B	Inverted operation	Puts the images in rows instead of columns.
C	Dropped element	Adds $T(e_1)$ and $T(e_2)$ without the weights 3 and 2.
D	Correct	$3(2, 1) + 2(-1, 3) = (6 - 2, 3 + 6) = (4, 9)$.

8. Answer A Concept 6

A population model in an ecology study uses $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. ~~The study tracks a bird species on a coastal island.~~ What are the eigenvalues of A ?

Target word: eigenvalues **Decides:** $\lambda^2 - 7\lambda + 10 = 0$.

A	Correct	Trace 7, det 10; $(\lambda - 5)(\lambda - 2) = 0$.
B	Inverted operation	Sign error in the factoring.
C	Wrong stem number	Those are the trace and determinant.
D	Confusion set	Diagonal entries are eigenvalues only for triangular matrices.

9. Answer C Concept 6 · transfer

A 2×2 matrix has eigenvalues 3 and -2. What is its determinant?

Target word: solve for det **Decides:** Determinant is the product of eigenvalues.

A	Inverted operation	Subtracts instead of multiplying.
B	Dropped element	Drops the negative sign.
C	Correct	$3 \times (-2) = -6$.
D	Confusion set	That is the trace, the sum.

10. Answer C Concept 7

A teacher fits a line to 20 data points from a class survey on sleep and grades. ~~The survey was done during finals week.~~ The system $Ax = b$ has no exact solution. Which equation gives the least-squares solution?

Target word: least-squares equation **Decides:** Normal equations.

A	Inverted operation	Wrong order of the transpose and missing A^T on b .
B	Absolute language	The stem says no exact solution exists.
C	Correct	Makes the error orthogonal to the column space of A .
D	Confusion set	A is 20×2 and has no inverse.

11. Answer D Concept 8

A 5×7 matrix from a supply-chain model has rank 3. ~~The model covers 14 warehouses.~~ What is the dimension of its null space?

Target word: nullity **Decides:** rank + nullity = number of columns.

A	Camouflage	Uses the 14 warehouses, which have nothing to do with the matrix size.
B	Confusion set	That is the rank.
C	Wrong stem number	Uses 5 rows instead of 7 columns.
D	Correct	$7 - 3 = 4$.

12. Answer D Concept 8 · transfer

A 6×8 matrix has a null space of dimension 5. What is its rank?

Target word: solve for rank **Decides:** rank = $8 - 5$.

A	Confusion set	That is the nullity.
B	Wrong stem number	Uses 6 rows instead of 8 columns.
C	Inverted operation	Adds instead of subtracting.
D	Correct	$8 - 5 = 3$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Matrix product dimensions	An $(m \times n)$ times $(n \times p)$ product is defined only when the inner sizes match, and the result is $m \times p$; AB and BA usually differ.	Using the inner dimensions as the result's size, or assuming BA exists because AB does.	Two matrix sizes and a question about a product.
2	Determinant properties	$\det(kA) = k^n \det(A)$ for an $n \times n$ matrix, $\det(AB) = \det(A)\det(B)$, $\det(A^{-1}) = 1/\det(A)$, and $\det(A) = 0$ means A is not invertible.	Thinking $\det(2A) = 2\det(A)$.	A scaled matrix, a product, an inverse, or a zero determinant.
3	Counting solutions of a linear system	After row reduction: a row $[0 \dots 0 \mid \text{nonzero}]$ means no solution; a consistent system with a free variable has infinitely many; no free variables means exactly one.	Reading a row of all zeros as no solution.	A reduced augmented matrix.
4	Linear independence and dimension	In $\mathbb{R}^n$, more than n vectors must be linearly dependent, and fewer than n vectors cannot span $\mathbb{R}^n$.	Thinking vectors are independent whenever none is a multiple of another.	A count of vectors compared with the dimension of the space.
5	Matrix of a linear transformation	The columns of the standard matrix are the images of the standard basis vectors; a linear map sends 0 to 0 .	Writing the images as rows, or adding them.	$T(e_1)$ and $T(e_2)$ given, and T of another vector asked.
6	Eigenvalues, trace, and determinant	Eigenvalues solve $\det(A - \lambda I) = 0$; their sum is the trace and their product is the determinant.	Reading eigenvalues off the diagonal of a matrix that is not triangular.	A 2×2 matrix and a question about eigenvalues, trace, or determinant.
7	Least squares	When $Ax = b$ has no solution, the least-squares solution solves $A^T A x = A^T b$, making the error vector orthogonal to the column space of A .	Trying to invert a non-square A or insisting $Ax = b$ be solved exactly.	More equations than unknowns, such as fitting a line to data points.
8	Rank-nullity theorem	For an $m \times n$ matrix, $\text{rank} + \text{nullity} = n$, the number of columns.	Using the number of rows instead of columns.	A matrix size and a rank or nullity, with the other asked.

Course scope (public description): Linear Algebra delves into the study of linear systems, matrices, and vector spaces, forming a critical foundation for advanced mathematics. Students will begin by mastering matrix operations, including solving linear systems and finding determinants and inverses. As they progress, they will explore vector spaces, linear transformations, and inner product spaces, gaining the skills to work with higher-dimensional systems. The course also covers advanced topics such as eigenvalues, eigenvectors, diagonalization, and sophisticated techniques like the least squares method, linear programming, and singular value decomposition. This course is essential for those preparing for Multivariable Calculus and other advanced mathematical studies.